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Cryoablation Reduces Opioid Consumption and Length of Stay After Pulmonary Metastasectomy

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ABSTRACT

Introduction: Intraoperative cryoablation of intercostal thoracic nerves is gaining popularity as a technique that decreases postoperative pain in thoracic surgery. Our study evaluates the efficacy and safety of cryoablation in pain management of pediatric cancer patients undergoing thoracotomy.

Methods: We reviewed cancer patients undergoing thoracotomies for pulmonary metastasis resection at our children's hospital from 2017 to 2023. Patients who received cryoablation were compared to those who did not. Our primary outcomes were self-reported postoperative pain scores (from 0 to 10) and opioid consumption, measured as oral morphine equivalent per kilogram.

Results: Thirty eight procedures were performed in 17 patients, of which 11 (64.7%) were males. Cryoablation was used in 14 (32.4%) procedures, while it was not in 24 (67.6%). Median age (17 y in both groups, $P = 0.84$) and length of surgery (300 cryoablation *versus* 282 no cryoablation, $P = 0.65$) were similar between the groups. Patients treated with cryoablation had a shorter hospital stay compared to those who did not (3.0 *versus* 4.5 d, respectively, $P = 0.04$) and received a lower total dose of opioids (2.2 oral morphine equivalent per kilogram *versus* 14.4, $P = 0.004$). No significant difference was noted in daily pain scores between the two groups (3.8 cryoablation *versus* 3.9 no cryoablation, $P = 0.93$). There was no difference in rates of readmissions between the cryoablation and no-cryoablation groups (14.3% *versus* 8.3%, $P = 0.55$).

Conclusions: Our study suggests that cryoablation of the thoracic nerves during a thoracotomy is associated with reduced opiate consumption and shorter hospital stay. Cryoablation appears to be a promising technique for pain management in this patient population.

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Introduction

Thoracotomy, for any indication, is known to be associated with significant postoperative pain in both adults and children. This often leads to challenges in effectively managing pain in these patients, including increased opioid use, ongoing discomfort, difficulty with mobilization and ambulation postoperatively, and prolonged hospital stays.[1-5]

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Management of pain after thoracotomy may require a variety of modalities including systemic opiates, regional analgesics, and new oral and parenteral agents.[6,7,8]

Intercostal nerve cryoablation, first developed in 1961 by Dr Irving S. Cooper, involves using extremely cold temperatures to freeze specific nerves which helps reduce pain. This is achieved using a special device that employs cold gases such as nitrous oxide or argon to create ice around the target area. The cold essentially stops pain signals from traveling, providing relief to the patient.[9] This technique has been used in different procedures, most commonly in Nuss procedure for pectus excavatum repair.[8,10-15] It has also been used for post-thoracotomy pain management[16-18] and in abdominal aortic aneurysms repair.[19] Studies have shown that it can help reduce pain and the need for opioid pain medication in pediatric cancer patients. However, its impact on hospital stays is not as clear. Resection of pulmonary metastasis in children often requires open thoracotomy. Our study aims to evaluate the efficacy and safety of cryoablation in the pain management of cancer patients undergoing thoracotomy for pulmonary metastasectomy using our institutional database.

Materials and Methods

We conducted a retrospective cohort study of all patients aged less than 25 y who underwent open thoracotomy procedures for the excision of pulmonary metastases from January 2017 to August 2023 at a large single academic institution and tertiary referral center. For patients who receive more than one surgery, each procedure was counted separately for the analysis. This project received full approval from the Johns Hopkins International Review Board (IRB00358188). Individual patient consent was not sought, as the study operated under an Institutional Review Board–approved waiver of consent in compliance with the Health Insurance Portability and Accountability Act.

Intercostal nerve cryoablation

During the intraoperative procedure, cryoablation was conducted using a CryoSPHERE probe (manufactured by AtriCure, Mason, OH). The decision to perform cryoablation was made based on the clinical judgment of the primary attending surgeon. In most cases, the thoracotomy was performed at the fifth intercostal space. The cryoablation probe was applied to the intercostal bundle posteriorly at the fifth intercostal level, as well as two intercostal spaces above and two below. In certain situations, additional intercostal bundles were performed inferiorly to ensure coverage of the chest tube site. Each nerve was subjected to cryoablation for a duration of 2 min. The cryoablation procedure was conducted at the beginning of the surgery, preceding the resection of metastatic lesions. To note that no patient declined cryoablation throughout the course of our study. Moreover, cryoablation was introduced at our institution in 2019. Therefore, all patients treated before 2019 did not receive cryoablation, while not all patients after 2019 received cryoablation.

Outcomes

The primary outcome of interest was inpatient postoperative oral morphine equivalent per kilogram (OME/Kg) usage. OMEs were calculated using conversion factors derived from international sources.[20] Secondary outcomes included patients' self-reported postoperative pain scores on a scale of 0 (indicating no pain) to 10 (indicating excruciating pain) at 4-h intervals. We also examined postoperative length of stay (LOS), defined as the time from surgery to discharge order, and monitored hospital readmissions as well as emergency department visits related to the surgical procedures. Notably, our OME calculations excluded intrathecal medications from single intraoperative injections, epidural catheter infusion, and intraoperative Fentanyl use.

Statistical analysis

We conducted a comparative analysis to evaluate differences in characteristics and outcomes between patients who underwent thoracotomy with cryoablation and those who underwent thoracotomy without cryoablation. Categorical variables, presented as frequency and percentage, were evaluated with Pearson's chi-squared or Fisher's exact tests. For continuous variables, normality was assessed using the Shapiro–Wilk test. Non-normally distributed variables were compared using the Mann–Whitney *U*-test, presented as median (interquartile range), while normally distributed variables were analyzed using the student's *t*-test and presented as mean ± standard deviation. To explore the association between cryoablation and total OME/kg administered during the hospital stay, multivariate linear regression was performed. The final model incorporated sex and age as potential confounding factors. Weight or body mass index was not included in the model, as weight was already accounted for in the OME/Kg calculation. A P value of less than 0.05 was considered statistically significant. All statistical analyses were performed using R Statistical software version 4.2.2 (2022-10-31 ucrt, "Innocent and Trusting").

Results

A total of 17 individual patients were included in the study, with a total of 38 thoracotomies performed among them. Twenty four of these (63.2%) were performed without Cryoablation and 14 (36.8%) with Cryoablation. Of the 17 individual patients, 11 (64.7%) were male. The median age at surgery was 17 y (interquartile range 15.0-18.5), with no difference found between the two groups ($P = 0.84$). While the youngest patient receiving cryoablation was aged 14 y, the overall age range in the cohort extended from 11 to 25 y. There was no significant difference in sex ($P = 0.45$), age ($P = 0.84$), weight ($P = 0.51$), height ($P = 0.59$), and body mass index ($P = 0.61$) between the two groups.

Among these patients, one patient had a malignant germ cell tumor and underwent a single surgery, another patient had a malignant peripheral nerve sheath tumor of the upper extremity and required two separate procedures, whereas the

Table 1 – Demographic of patients undergoing thoracotomy for pulmonary metastasectomy by cryoablation *versus* no cryoablation.

Patient characteristics	Total $n = 38$ (%)	Cryoablation $n = 14$ (%)	No cryoablation $n = 24$ (%)	*P* value
Diagnosis				**0.03**
Germ cell tumor	2 (5.3)	2 (14.3)	0 (0)	
MPNST	1 (2.6)	1 (7.1)	0 (0)	
Osteosarcoma	35 (92.1)	11 (78.6)	24 (100)	
Sex				0.45
Female	10 (26.3)	5 (35.7)	5 (20.8)	
Male	28 (73.7)	9 (64.3)	19 (79.2)	
Age (Y)	17 [15-18.5]	17 [14-18.5]	17 [16-18.5]	0.84
Height (cm)	168 [161-179]	163 [161-176]	170 [164-178]	0.51
Weight (Kg)	57.6 [50.1-67.0]	53.0 [48.6-64.0]	60.7 [52.8-68.2]	0.59
BMI (Kg/m^2)	20.0 [17.6-24.4]	18.0 [17.2-24.4]	20.5 [18.8-24.0]	0.61
Additional methods of pain control				
Epidural	26 (68.4)	9 (64.3)	15 (62.5)	0.91
Ketamine	8 (21.1)	3 (21.4)	5 (20.8)	0.97
Peripheral nerve block	2 (5.3)	1 (7.1)	1 (4.2)	0.69

BMI = body mass index; IQR = interquartile range; PNST = peripheral neural sheath tumors.
P values in bold indicate statistical significance ($P < 0.05$).

remaining 15 individuals were diagnosed with osteosarcoma, with 12 of them requiring multiple surgeries.

There were no significant differences in the use of additional postoperative analgesia methods between the cryoablation and no-cryoablation groups. These methods included epidural catheter placement ($P = 0.91$), ketamine ($P = 0.97$), and peripheral nerve block ($P = 0.69$) (Table 1).

Primary outcome

Patients who underwent cryoablation consumed significantly less opioid during their hospitalization compared to patients who did not (median 1.1 [0.2-4.2] OME/kg *versus* 9.5 [3.5-20.8] OME/kg, respectively, $P = 0.0002$) (Table 2, Fig. 1). Notably, when comparing opioid consumption per postoperative day (POD), patients with cryoablation exhibited notably lower daily opioid consumption on each POD from 0 through 6 compared to those without cryoablation ($P < 0.05$) (Fig. 2, Appendix 1). On multivariate analysis controlling for age, and sex, cryoablation demonstrated a significant reduction in opioid consumption compared to no cryoablation, with a substantial decrease of 11.2 OME/kg (95% confidence interval −18.9 to −3.6, $P = 0.007$) (Table 3).

Secondary outcomes

The maximum patient-reported and nurse-reported daily postoperative pain scores ranged from 0 to 10 in both patient groups. The median daily maximum pain score reported by cryoablation patients (7.0 [6.0-7.8]) did not significantly differ from no-cryoablation patients (7.5 [5.0-9.0]) overall ($P = 0.20$) (Table 2, Fig. 1). Furthermore, when evaluated by POD, there

Table 2 – Outcomes of patients undergoing thoracotomy for pulmonary metastasectomy by cryoablation *versus* no cryoablation.

Outcomes	Total $n = 38$	Cryoablation $n = 14$	No cryoablation $n = 24$	*P* value
EBL (mL)	50 [10-100]	50 [35-175]	23 [10-81]	0.55
LOS (d), median [IQR]	4.0 [3.0-6.0]	3.0 [3.0-4.8]	4.5 [3.8-6.2]	**0.02**
Length of surgery (min)	288 ± 123	300 ± 120	282 ± 127	0.65
Persistent air leak	9 (23.7)	2 (14.3)	7 (29.3)	0.26
Unplanned readmission, *n* (%)	4 (10.5)	2 (14.3)	2 (8.3)	0.55
Total OME/Kg given	4.7 [1.3-11.7]	1.1 [0.2-4.2]	9.5 [3.5-20.8]	**0.0002**
Maximum pain score	7.0 [6.0-8.8]	7 [6.0-7.8]	7.5 [5.0-9.0]	0.60

IQR = interquartile range; LOS = length of stay; EBL = estimated blood loss.
All continuous variables are shown as mean ± standard deviation unless otherwise specified.
P values in bold indicate statistical significance ($P < 0.05$).

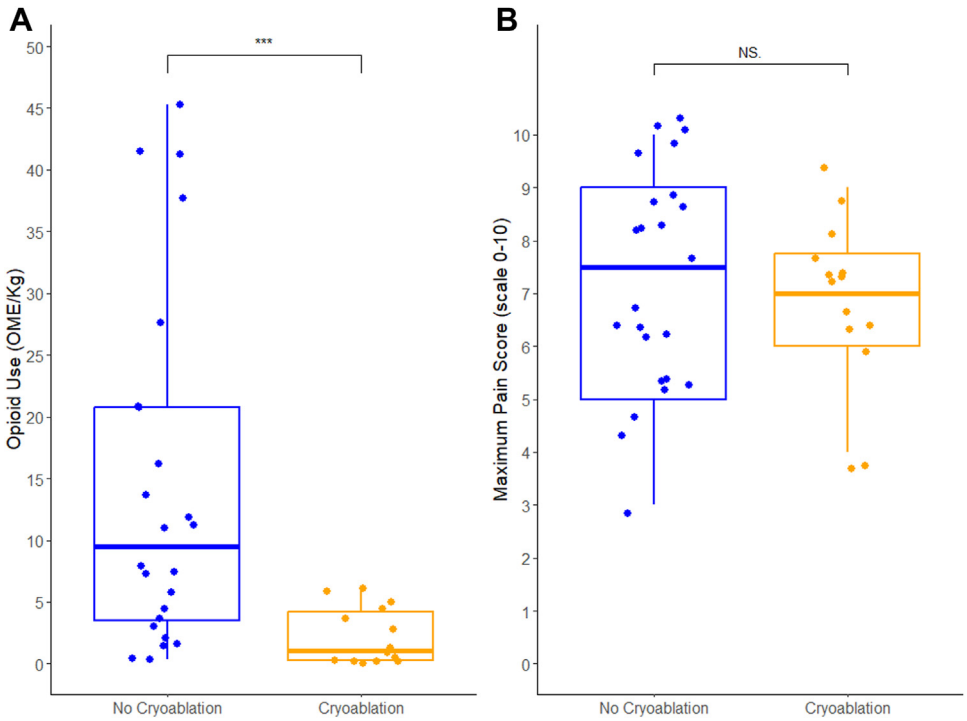

Fig. 1 – Comparative analysis of total opioid use (OME/Kg) (A) and average pain score (B) during the entire hospital stay between patients with (yellow) and without (blue) cryoablation. * indicates a *P* value < 0.001. NS = not statistically significant.**

was no significant difference in the maximum reported pain score on any POD (Fig. 3, Appendix 2).

Patients treated with cryoablation experienced a significantly shorter postoperative LOS compared to those who did not (median 3.0 d *versus* 4.5 d, respectively, $P = 0.03$). The mean duration of surgery in patients who received cryoablation was 300 min (±120) compared to 282 min (±127) for those without cryoablation; however, this difference was not statistically significant ($P = 0.65$). No intraoperative complications were reported, and estimated blood loss during the

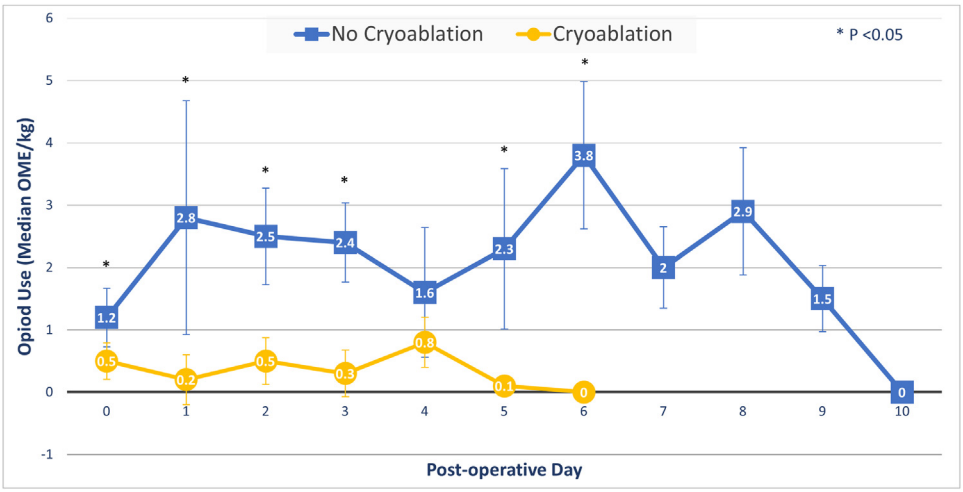

Fig. 2 – Median OME/kg per postoperative day in children who received cryoablation (yellow circles) *versus* no cryoablation (blue squares). Data shown as median oral morphine equivalent per kilogram (OME/Kg) per postoperative day. Mann-Whitney *U*-test was used to compare the treatment groups. * indicates *P* value < 0.05 which was considered statistically significant.

Table 3 – Multivariate linear regression analysis for total OME/kg given during the hospital stay, controlling for age, sex, and BMI.

Variables	β coefficient	95% confidence interval	P value
Cryoablation	−11.2	−18.9 to −3.6	**0.007**
Age	−0.2	−1.5 to 1.1	0.79
Male sex	5.6	−3.6 to 14.9	0.23

BMI = body mass index; OME, oral morphine equivalent.
P values in bold indicate statistical significance ($P < 0.05$).

surgery did not significantly differ between the two groups (50 mL for cryoablation *versus* 23 mL for no cryoablation, $P = 0.55$). There was no significant difference in the development of persistent air leaks between the two groups (14.3% for cryoablation *versus* 29.3% for no cryoablation, $P = 0.26$) (Table 2).

In terms of readmission, two patients (14.3%) who received cryoablation were readmitted, while two (8.3%) were readmitted in the no-cryoablation group ($P = 0.55$). The reasons for readmission were unrelated to pain (Table 2).

Discussion

Our study findings showed that intraoperative cryoablation of thoracic nerves during thoracotomy for pulmonary metastasectomy was associated with reduced postoperative opioid consumption and shorter hospital stays, without increasing the risks of complications, readmissions, or reoperations. This study highlights the potential of cryoablation as a promising technique for pain management in this patient population. Our findings contribute to the existing body of literature on the efficacy and versatility of cryoablation by demonstrating its effectiveness in these procedures among pediatric patients with metastatic lung lesions.[21,22]

Young patients with metastatic lung lesions frequently require thoracotomy for lesion excision, and managing pain in this patient population is a multifaceted challenge due to the invasive nature of the surgery. Interestingly, our study did not demonstrate a difference in reported pain scores between the two groups; however, the reduction in postoperative opioid use in the cryoablation group to achieve the same reported pain scores supports its effectiveness in alleviating pain. Furthermore, given the current opioid epidemic, minimizing opioid prescriptions, especially in the pediatric and young adult population, is crucial in mitigating the side effects and addictive potential of these medications. Cryoablation have been consistently shown to decrease pain and opioid consumption in the setting of the Nuss procedure for pectus excavatum repair.[13,14,23] In the setting of pediatric and adolescent cancer patients undergoing thoracotomy, Chen *et al* and McElhinney *et al* demonstrated a similar reduction in postoperative inpatient opioid use. In addition, they showed that patients who received cryoablation were prescribed less opioids on discharge.[21,22]

A noteworthy outcome of our study was the significantly shorter postoperative LOS for patients who received cryoablation compared to those who did not (3.0 d *versus* 4.5 d, respectively; $P = 0.03$). This reduction by a day and a half is particularly relevant, considering that previous studies in Nuss procedures have consistently revealed similar findings. However, both Chen *et al* and McElhinney *et al*'s investigations into cryoablation during thoracotomy did not yield this same result:[21,22] Postoperative LOS did not differ between patients who received cryoablation and those who did not. In our cohort, we attribute the observed shorter LOS in the cryoablation group to superior pain management. Nonetheless, several contributing factors may lead to this difference, including inherent variations in patient populations, chest tube management, degree of pulmonary resection, and postoperative recovery protocols. Our cohort exhibited uniformity in conjunctive pain management methods, including the use of epidurals and peripheral nerve blocks, among patients who

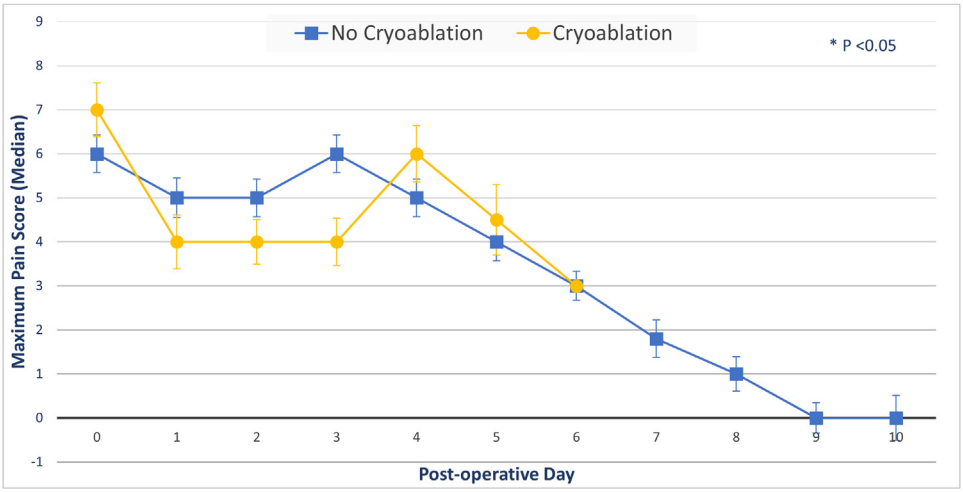

Fig. 3 – Median maximum pain scores (scale 0-10) per postoperative day in children who received cryoablation (yellow circles) *versus* no cryoablation (blue squares). Data shown as median maximum pain score on a scale of 0-10 per postoperative day. Mann-Whitney *U*-test was used to compare the treatment groups. There was no statistically significant difference between the groups.

underwent cryoablation and those who did not. Additionally, chest tubes were maintained until the day of discharge or the preceding day for all patients. Despite the lack of statistical significance, patients not receiving cryoablation developed more persistent air leaks requiring prolonged chest tube placement, potentially contributing to the observed difference in LOS.

Furthermore, postoperative management variations, especially in the context of the extent of lung resections performed, may elevate the risk of prolonged air leaks and extended hospital stays. Notably, our institution currently lacks an Enhanced Recovery After Surgery protocol. Nevertheless, the establishment of a comprehensive and standardized perioperative care approach, through an Enhanced Recovery After Surgery protocol, is imperative to systematically address and optimize various aspects of perioperative care, potentially further improving outcomes and reducing hospitalization durations.[24-26] Further studies involving a larger patient cohort are needed to establish the generalizability of these findings. These findings carry significant implications as it not only reduces the burden on healthcare facilities but also enhances patient quality of life and mental well-being by minimizing their hospital stays, thus emphasizing its importance in promoting better patient outcomes.[27-29] Moreover, in addition to shortening LOS, research has shown that cryoablation may also contribute to reducing total hospital costs, as observed in patients with pectus excavatum undergoing Nuss procedure.[23,30,31] Future studies would help validate these findings in our patient population.

Our study is not without limitations. First, our study design is retrospective in nature, potentially introducing biases and limiting our ability to establish causal relationships. Second, our study is constrained by a relatively small sample size. This limitation inherently affects the generalizability of our findings. Additionally, the introduction of cryoablation in 2019 increases the potential of time bias. Furthermore, while both groups received similar nonopioid pain management strategies, including the use of nonsteroidal anti-inflammatory drugs, epidural analgesia, and ketamine, there were some variations in individualized treatment plans that could not be controlled for. Further research, ideally in the form of prospective studies with larger and more diverse patient populations, is necessary to establish more robust and widely applicable conclusions.

In conclusion, our study findings support the efficacy of cryoablation in reducing both opioid utilization and the length of hospital stays in children who have undergone open thoracotomy for the excision of lung metastases. This finding underscores the potential of cryoablation as a promising and effective analgesic technique in this population. Therefore, cryoablation warrants careful consideration and incorporation into the comprehensive management strategy for these patients to enhance their overall quality of care and postoperative recovery.

Supplementary Materials

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jss.2024.01.044.

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CRediT authorship contribution statement

Charbel Chidiac: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Kristin Wharton:** Writing – original draft, Formal analysis. **Alejandro V. Garcia:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **Daniel S. Rhee:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization.

REFERENCES

1. Danielsen AV, Andreasen JJ, Dinesen B, et al. Chronic post-thoracotomy pain after lung cancer surgery: a prospective study of preoperative risk factors. *Scand J Pain*. 2023;23:501–510.
2. Katz J, Jackson M, Kavanagh BP, Sandler AN. Acute pain after thoracic surgery predicts long-term post-thoracotomy pain. *Clin J Pain*. 1996;12:50.
3. Kodia K, Stephens-McDonnough JA, Alnajar A, Villamizar NR, Nguyen DM. Implementation of an enhanced recovery after thoracic surgery care pathway for thoracotomy patients—achieving better pain control with less (schedule II) opioid utilization. *J Thorac Dis*. 2021;13:3948–3959.
4. Macrae WA. Chronic pain after surgery. *Br J Anaesth*. 2001;87:88–98.
5. Marshall K, McLaughlin K. Pain management in thoracic surgery. *Thorac Surg Clin*. 2020;30:339–346.
6. Effective pain management and improvements in patients' outcomes and satisfaction | critical care nurse | American association of critical-care nurses. Available at: https://aacnjournals.org/ccnonline/article-abstract/35/3/33/3485/Effective-Pain-Management-and-Improvements-in. Accessed October 31, 2023.
7. Landreneau RJ, Hazelrigg SR, Mack MJ, et al. Postoperative pain-related morbidity: video-assisted thoracic surgery versus thoracotomy. *Ann Thorac Surg*. 1993;56:1285–1289.
8. Soto RG, Fu ES. Acute pain management for patients undergoing thoracotomy. *Ann Thorac Surg*. 2003;75:1349–1357.
9. Evans PJ. Cryoanalgesia. The application of low temperatures to nerves to produce anaesthesia or analgesia. *Anaesthesia*. 1981;36:1003–1013.
10. Daemen JHT, de Loos ER, Vissers YLJ, Bakens MJAM, Maessen JG, Hulsewé KWE. Intercostal nerve cryoablation versus thoracic epidural for postoperative analgesia following pectus excavatum repair: a systematic review and meta-analysis. *Interact Cardiovasc Thorac Surg*. 2020;31:486–498.
11. Dekonenko C, Dorman RM, Duran Y, et al. Postoperative pain control modalities for pectus excavatum repair: a prospective observational study of cryoablation compared to results of a randomized trial of epidural vs patient-controlled analgesia. *J Pediatr Surg*. 2020;55:1444–1447.

12. Graves CE, Moyer J, Zobel MJ, et al. Intraoperative intercostal nerve cryoablation during the Nuss procedure reduces length of stay and opioid requirement: a randomized clinical trial. *J Pediatr Surg*. 2019;54:2250–2256.
13. Harbaugh CM, Johnson KN, Kein CE, et al. Comparing outcomes with thoracic epidural and intercostal nerve cryoablation after Nuss procedure. *J Surg Res*. 2018;231:217–223.
14. Keller BA, Kabagambe SK, Becker JC, et al. Intercostal nerve cryoablation versus thoracic epidural catheters for postoperative analgesia following pectus excavatum repair: preliminary outcomes in twenty-six cryoablation patients. *J Pediatr Surg*. 2016;51:2033–2038.
15. Sun RC, Mehl SC, Anbarasu CR, et al. Intercostal cryoablation during Nuss procedure: a large volume single surgeon's experience and outcomes. *J Pediatr Surg*. 2021;56:2229–2234.
16. Koethe Y, Mannes AJ, Wood BJ. Image-guided nerve cryoablation for post-thoracotomy pain syndrome. *Cardiovasc Intervent Radiol*. 2014;37:843–846.
17. Roberts D, Pizzarelli G, Lepore V, Al-Khaja N, Belboul A, Dernevik L. Reduction of post-thoracotomy pain by cryotherapy of intercostal nerves. *Scand J Thorac Cardiovasc Surg*. 1988;22:127–130.
18. Yasin J, Thimmappa N, Kaifi JT, et al. CT-guided cryoablation for post-thoracotomy pain syndrome: a retrospective analysis. *Diagn Interv Radiol*. 2020;26:53–57.
19. Clemence J, Malik A, Farhat L, et al. Cryoablation of intercostal nerves decreased narcotic usage after thoracic or thoracoabdominal aortic aneurysm repair. *Semin Thorac Cardiovasc Surg*. 2020;32:404–412.
20. Nielsen S, Degenhardt L, Hoban B, Gisev N. A synthesis of oral morphine equivalents (OME) for opioid utilisation studies. *Pharmacoepidemiol Drug Saf*. 2016;25:733–737.
21. Chen SY, Mack SJ, Stein JE, Kelley-Quon LI, Kim ES. Intercostal nerve cryoablation is associated with reduced opioid use in pediatric oncology patients. *J Surg Res*. 2023;283:377–384.
22. McElhinney KL, Zeineddin S, Ahle SL, Goldstein SD, Lautz TB. Intercostal nerve cryoablation reduces opioid utilization after thoracotomy in children with cancer. *Pediatr Blood Cancer*. 2024;71:e30722.
23. Mehl SC, Sun RC, Anbarasu CR, et al. Association of intercostal nerve cryoablation during Nuss procedure with complications and costs. *Ann Thorac Surg*. 2023;116:803–809.
24. Wharton K, Chun Y, Hunsberger J, Jelin E, Garcia A, Stewart D. Successful use of an enhanced recovery after surgery (ERAS) pathway to improve outcomes following the Nuss procedure for pectus excavatum. *J Pediatr Surg*. 2020;55:1065–1071.
25. Echeverria-Villalobos M, Stoicea N, Todeschini AB, et al. Enhanced recovery after surgery (ERAS) : a perspective review of postoperative pain management under ERAS pathways and its role on opioid crisis in the United States. *Clin J Pain*. 2020;36:219–226.
26. Sánchez-Iglesias JL, Gómez-Hidalgo NR, Pérez-Benavente A, et al. Importance of enhanced recovery after surgery (ERAS) protocol compliance for length of stay in ovarian cancer surgery. *Ann Surg Oncol*. 2021;28:8979–8986.
27. Decision support for hospital bed management using adaptable individual length of stay estimations and shared resources | BMC Medical Informatics and Decision Making | Full Text. Available at: https://bmcmedinformdecismak.biomedcentral.com/articles/10.1186/1472-6947-13-3. Accessed November 2, 2023.
28. The impact of inpatient bed capacity on length of stay | the European Journal of Health Economics. Available at: https://link.springer.com/article/10.1007/s10198-021-01373-2. Accessed November 2, 2023.
29. Freire T, Ferreira G, Pereira A. How happy children and adolescents feel during hospitalization in a pediatric unit? *Psic Saúde & Doenças*. 2021;22:218–229.
30. Aiken TJ, Stahl CC, Lemaster D, et al. Intercostal nerve cryoablation is associated with lower hospital cost during minimally invasive Nuss procedure for pectus excavatum. *J Pediatr Surg*. 2021;56:1841–1845.
31. Cost and outcomes of intercostal nerve cryoablation versus thoracic epidural following the Nuss procedure - ScienceDirect. Available at: https://www.sciencedirect.com/science/article/abs/pii/S0022346822007849?casa_token=EpHlQFXYTuMAAAAA:7Y8dPyQ4lzU1IZLrRjA9-fnyVtFKfQjwr7y3-i-V4PFFB9Of4xczU02BHiFN1Bh_eKOa-8cnoA. Accessed October 31, 2023.